USD/TRY Exchange Rate Time Series Analysis

Structural Break, Stationarity and Forecasting Analysis with R

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1. Introduction

This study analyzes the USD/TRY exchange rate series using time series techniques in R. The analysis focuses on stationarity testing, differencing transformation, autocorrelation structure, structural break interpretation, and ARIMA-based forecasting.

The primary objective is to investigate the statistical behavior of the USD/TRY exchange rate and evaluate whether the series is suitable for forecasting applications.

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2. Libraries and Data Import

The following R libraries were used during the analysis:

```
> library(tseries)

> library(quantmod)

> library(ggplot2)

> library(strucchange)

> library(urca)

> library(forecast)
```

The USD/TRY exchange rate data was retrieved from Yahoo Finance using the quantmod package:

```
> getSymbols("USDTRY=X", src = "yahoo")
```

The closing prices were extracted and missing observations were removed:

```
> usd_close <- Cl(`USDTRY=X`)

> usd_close <- na.omit(usd_close)
```

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3. Visualization of the Original Series

The original USD/TRY exchange rate series was visualized to examine the long-term trend and volatility behavior.

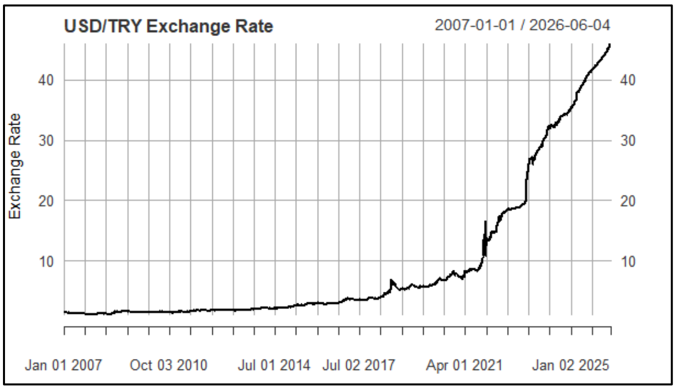
```
> plot(usd_close,

   main = "USD/TRY Exchange Rate",

   ylab = "Exchange Rate",

   xlab = "Time",

   col = "black")
```



The graph indicates a strong upward trend, especially after 2021. The increasing slope and volatility suggest that the series is non-stationary.

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4. Augmented Dickey-Fuller (ADF) Test – Level Series

To statistically evaluate stationarity, the Augmented Dickey-Fuller test was applied to the original series.

```
> adf_level <- adf.test(usd_close)

> adf_level
```

ADF Test Results

Dickey-Fuller = 2.1012

Lag order = 17

p-value = 0.99

Since the p-value is significantly greater than 0.05, the null hypothesis cannot be rejected. Therefore, the original USD/TRY series is considered non-stationary.

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5. First Difference Transformation

Because the original series was non-stationary, first differencing was applied to stabilize the mean and remove the trend component.

```
> usd_diff <- diff(usd_close)

> usd_diff <- na.omit(usd_diff)
```

The differenced series was then visualized:

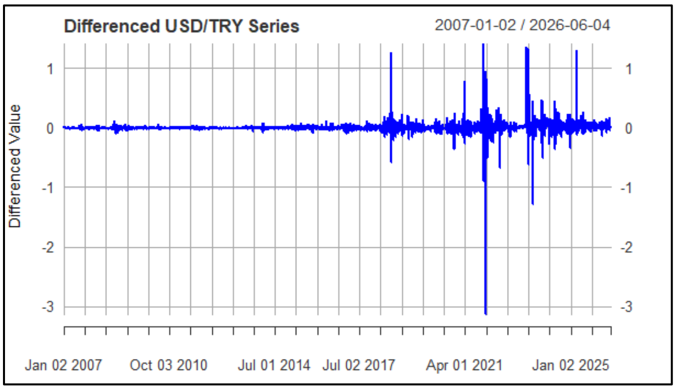
```
> plot(usd_diff,

    main = "Differenced USD/TRY Series",

    ylab = "Differenced Value",

    xlab = "Time",
```

col = "blue")



The differenced series fluctuates around zero and appears more stable compared to the original series.

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6. ADF Test – Differenced Series

The stationarity of the differenced series was tested again using the ADF test.

>adf_diff <- adf.test(usd_diff)

> adf_diff

ADF Test Results

Dickey-Fuller = -13.292

Lag order = 17

p-value = 0.01

Since the p-value is below 0.05, the null hypothesis is rejected. Thus, the differenced series is stationary.

This indicates that first differencing successfully stabilized the series.

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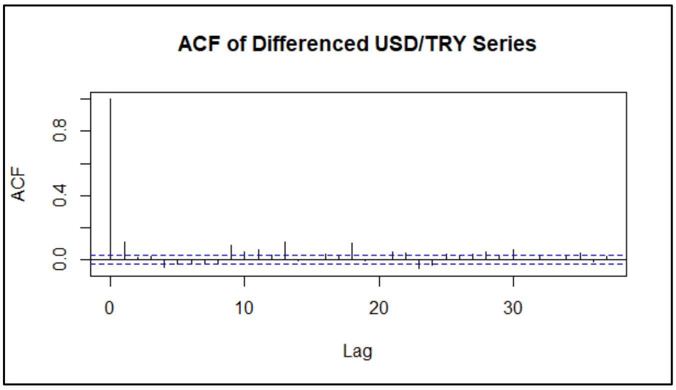
7. Autocorrelation Function (ACF)

The autocorrelation structure of the differenced series was examined using the ACF plot.
> acf(usd_diff,

main = "ACF of Differenced USD/TRY Series")



The ACF graph shows strong autocorrelation at the first lag followed by a rapid decay pattern, which is consistent with stationary time series behavior.

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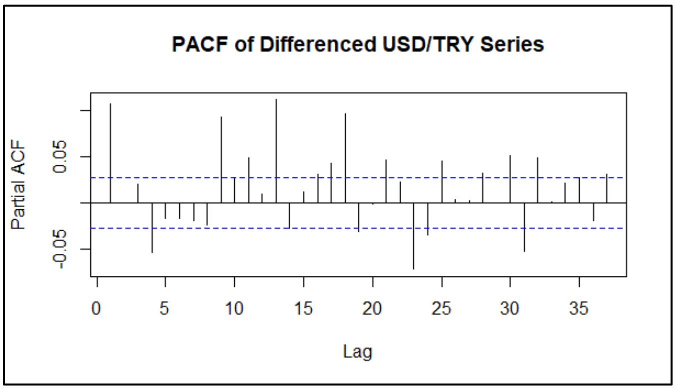
8. Partial Autocorrelation Function (PACF)

The PACF plot was analyzed to identify autoregressive dynamics within the series.

```
> pacf(usd_diff,
     main = "PACF of Differenced USD/TRY Series")
```



Several significant spikes are visible at early lags, indicating the presence of autoregressive components.

These findings support the use of ARIMA-based forecasting models.

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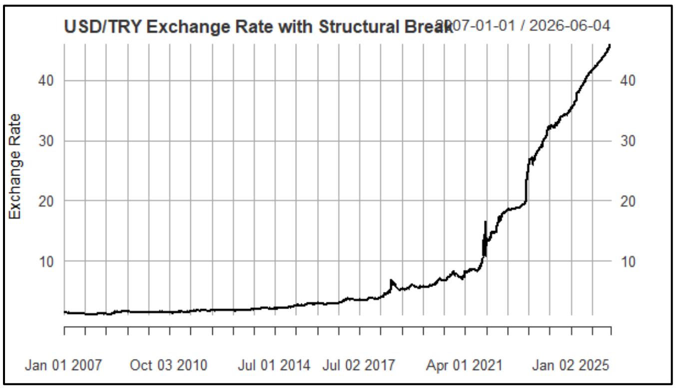
9. Structural Break Analysis

The USD/TRY series was evaluated for structural changes over time.

```
> plot(usd_close,
     main = "USD/TRY Exchange Rate with Structural Break",
     ylab = "Exchange Rate",
     xlab = "Time",
```

```
    col = "black")

> abline(v = as.Date("2021-12-01"),
       col = "red",
       lwd = 2,
       lty = 2)
```



A significant structural break was identified around December 2021. This period corresponds to major economic and monetary policy shifts in Türkiye, during which exchange rate volatility increased substantially.

Structural breaks are important because they change the statistical characteristics of the series and affect forecasting accuracy.

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10. ARIMA Model Selection

An automatic ARIMA model selection procedure was applied using the forecast package.

```
> arima_model <- auto.arima(usd_close)

> summary(arima_model)
```

Model Information Criteria

AIC = -8754.46

AICc = -8754.44

BIC = -8715.3

The relatively low information criteria values suggest that the selected ARIMA model provides a statistically acceptable fit.

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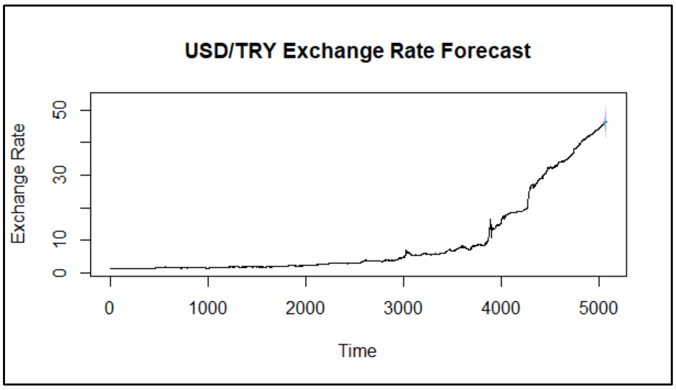
11. Forecasting Analysis

A 30-period forecast was generated using the selected ARIMA model.

```
> forecast_result <- forecast(arima_model, h = 30)
```

```
>plot(forecast_result,
    main = "USD/TRY Exchange Rate Forecast",
    ylab = "Exchange Rate",
    xlab = "Time")
```



The forecast results suggest that the upward trend in the USD/TRY exchange rate may continue in the short term.

The widening confidence intervals indicate increasing uncertainty in future exchange rate movements.

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12. Conclusion

This study analyzed the USD/TRY exchange rate using several time series analysis techniques in R.

The findings indicate that:

- the original series is non-stationary,
- first differencing successfully stabilizes the series,
- ACF and PACF structures support ARIMA modeling,
- and a major structural break occurred around December 2021.

The forecasting analysis further highlights the persistent volatility and uncertainty embedded in the exchange rate dynamics.

Overall, the study demonstrates the importance of stationarity testing, structural break interpretation, and ARIMA-based forecasting in financial time series analysis.